Course Name : Physics II Lab

Course Code : Phy 4242

Experiment No. : 07

Experiment Name : Verification of Maximum Power Transfer Theorem.

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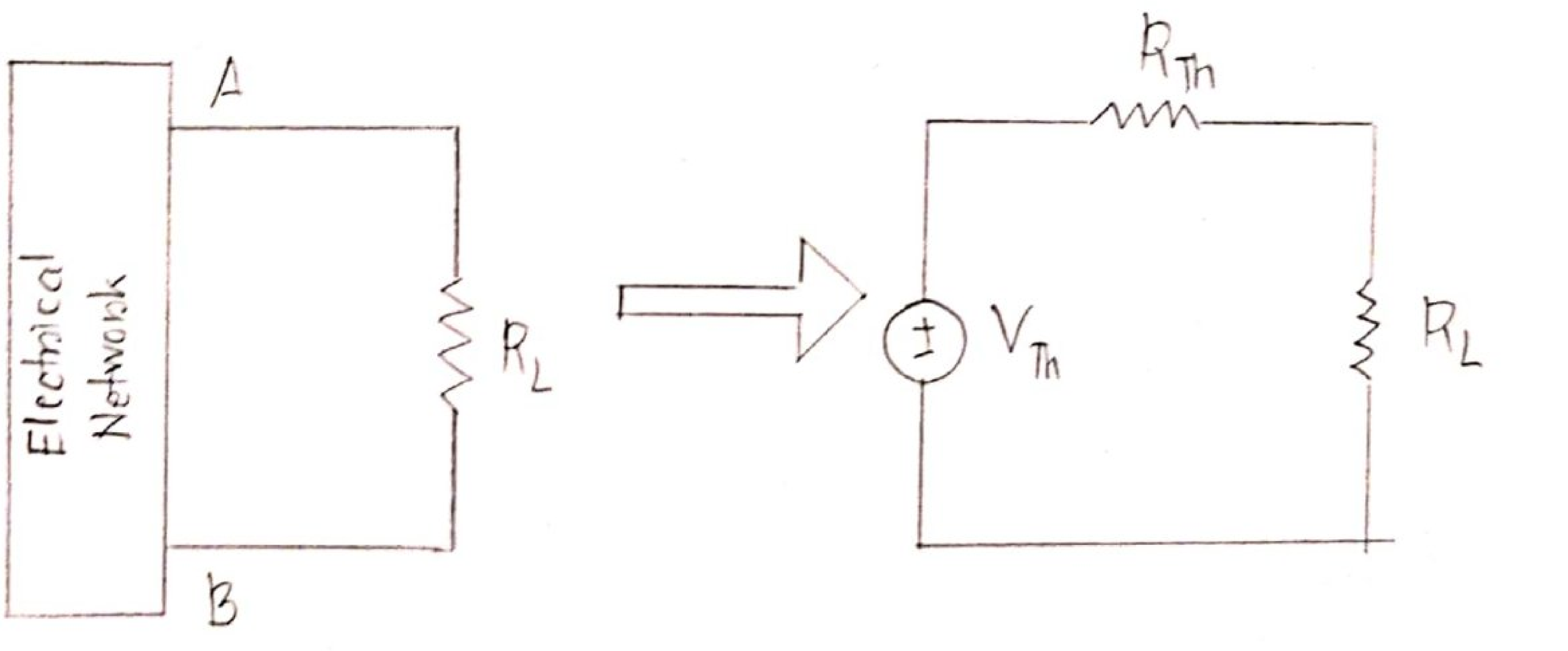
Objective :

The objective of this experiment is to verify maximum power transfer theorem.

Theory :

The maximum power transfer theorem states that,

"A resistive load receives maximum power when its total resistive value exactly equals the Thevenin's resistance of the network ~~as~~ seen by the load."



Maximum power is transferred when, $R_L = R_{Th}$.

Maximum power, $P_{max} = \dfrac{{V_{Th}}^2}{4R_{Th}}$

Circuit Diagram :

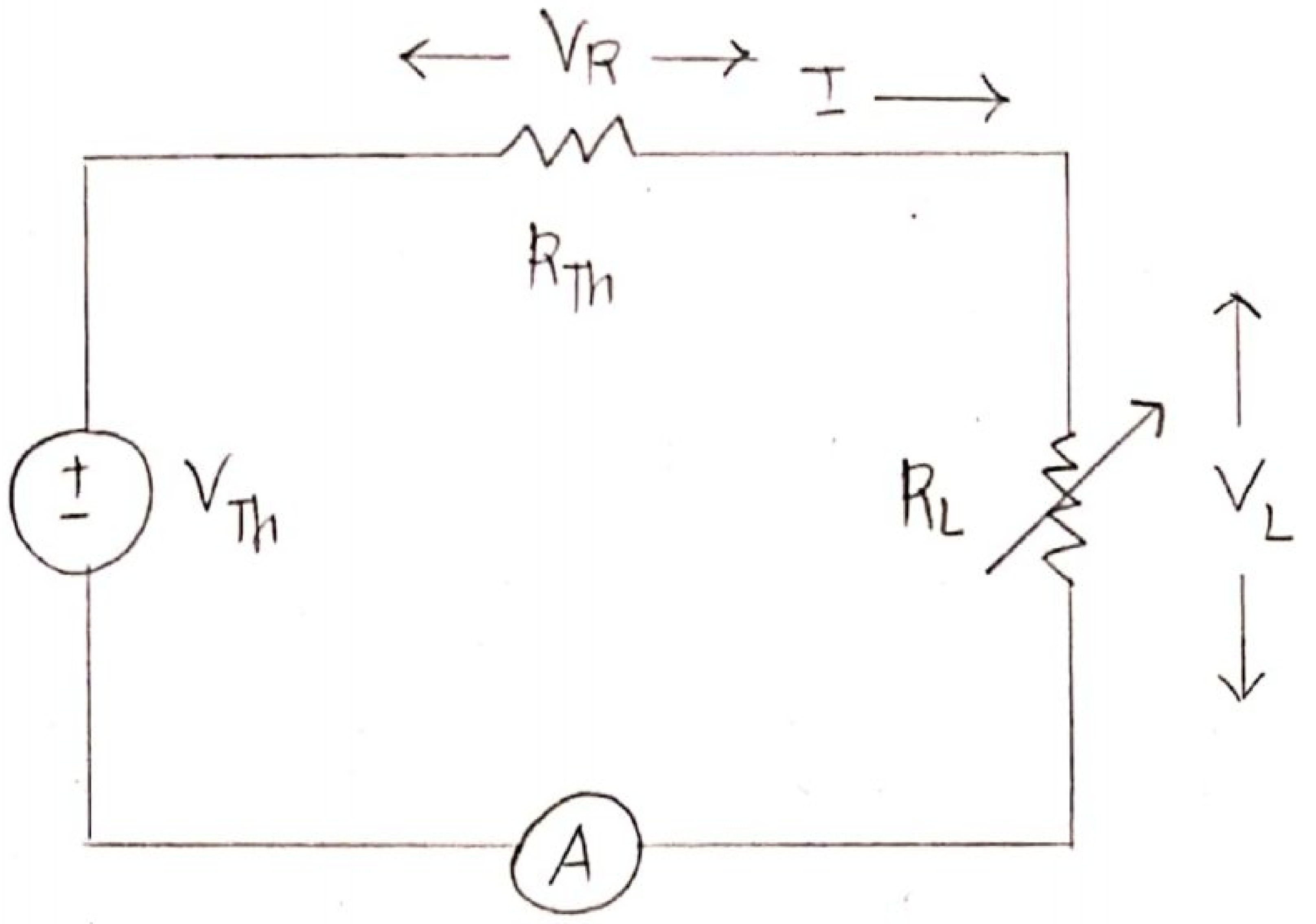


Figure : Circuit for experiment

Data Collection :

Obs. no.	V_{TH}	V_R (V)	V_L (V)	I (mA)	$P_{IN} = V_{TH} \cdot I$	$P_{OUT} = V_L \cdot I$	$R_L = V_L / I$	$R_{TH} = V_R / I$
1	15V	10.54	4.33	3.3	49.5 mV	14.29 mV	1.31	3.19
2	15V	9.39	6.12	2.9	43.5mV	17.75mV	2.11	3.24
3	15V	8.49	6.45	2.6	39 mV	16.77 mV	2.48	3.27
4	15V	7.76	7.17	2.4	36 mV	17.21 mV	2.99	3.23
5	15V	6.59	8.36	2	30 mV	16.72mV	4.18	3.295
6	15V	5.80	9.17	1.8	27 mV	16.51 mV	5.09	3.22
7	15V	5.06	9.82	1.6	24mV	15.71 mV	6.14	3.1625

Data Analysis:

From the collected data we can see that for the 4th observation is R_L is closest to R_{th}. So we can conclude that maximum power transfer can be achieved there.

For maximum power transfer,

$$\text{Efficiency}, \ \eta_4 = \frac{17.21}{36} \times 100 = 47.81\%$$

$$\text{Voltage regulation}, \ \eta_{v_4} = \frac{15. - 7.17}{7.17} \times 100 = 109.2\%$$

$$\text{Power loss}, \ P_{L4} = 36 - 17.21 = 18.79 \text{ mV}$$

For other observations,

Obs. 1,

$$\eta_1 = 28.87\%$$

$$\eta_{v_1} = 246.42\%$$

$$P_{L1} = 35.21 \text{ mV}$$

Obs. 2,

$$\eta_2 = 40.80\%$$

$$\eta_{v_2} = 145.09\%$$

$$P_{L2} = 25.75 \text{ mV}$$

Obs. 3,

$\eta_3 = 43\%$

$\eta_{v_3} = 132.56\%$

$P_{L_3} = 22.23 \text{ mV}$

Obs. 5,

$\eta_5 = 55.73\%$

$\eta_{v_5} = 79.426\%$

$P_{L_5} = 13.28 \text{ mV}$

Obs. 6,

$\eta_6 = 61.15\%$

$\eta_{v_6} = 63.58\%$

$P_{L_6} = 10.49 \text{ mV}$

Obs. 7,

$\eta_7 = 65.46\%$

$\eta_{v_7} = 52.75\%$

$P_{L_7} = 8.29 \text{ mV}$

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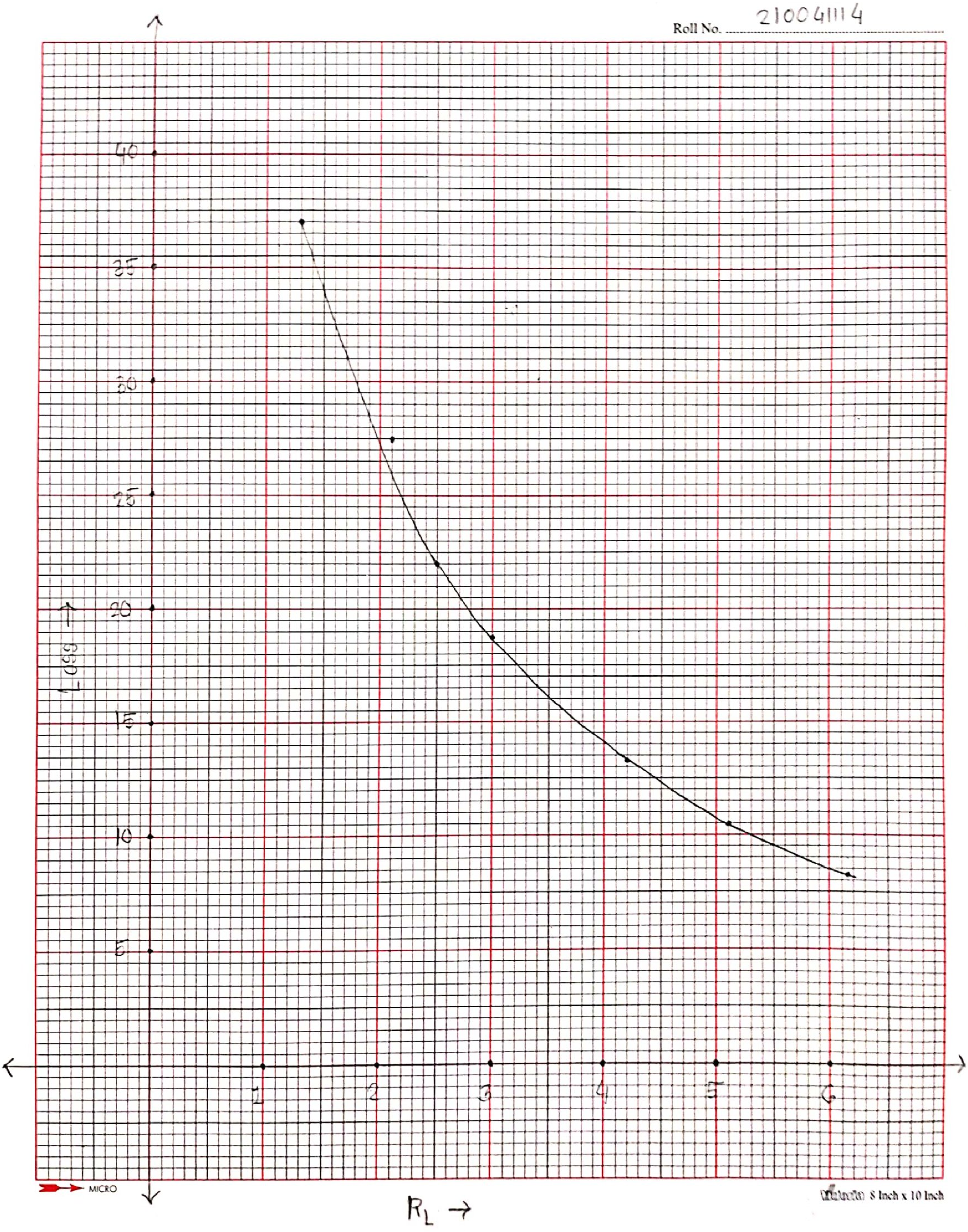

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8 Inch x 10 Inch

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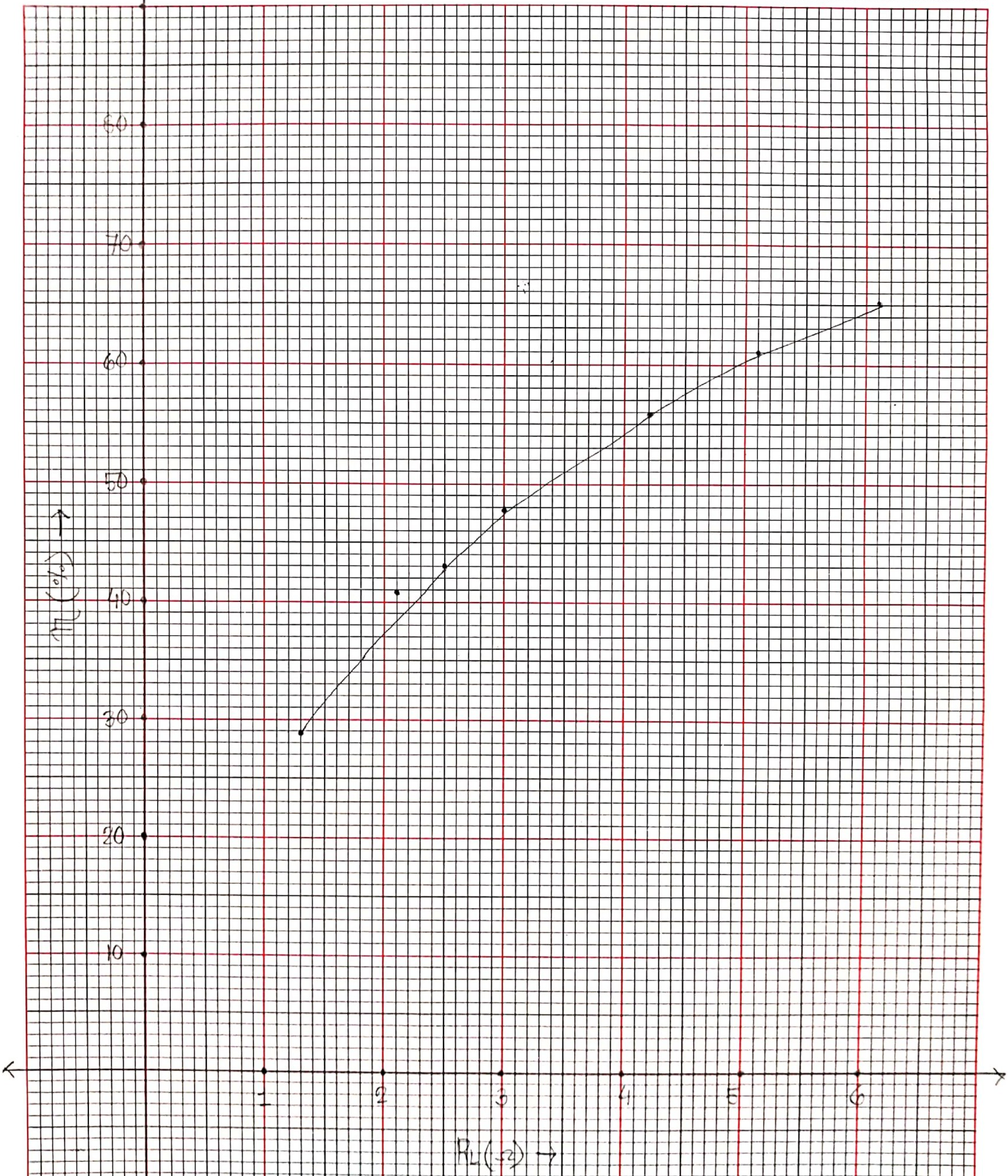

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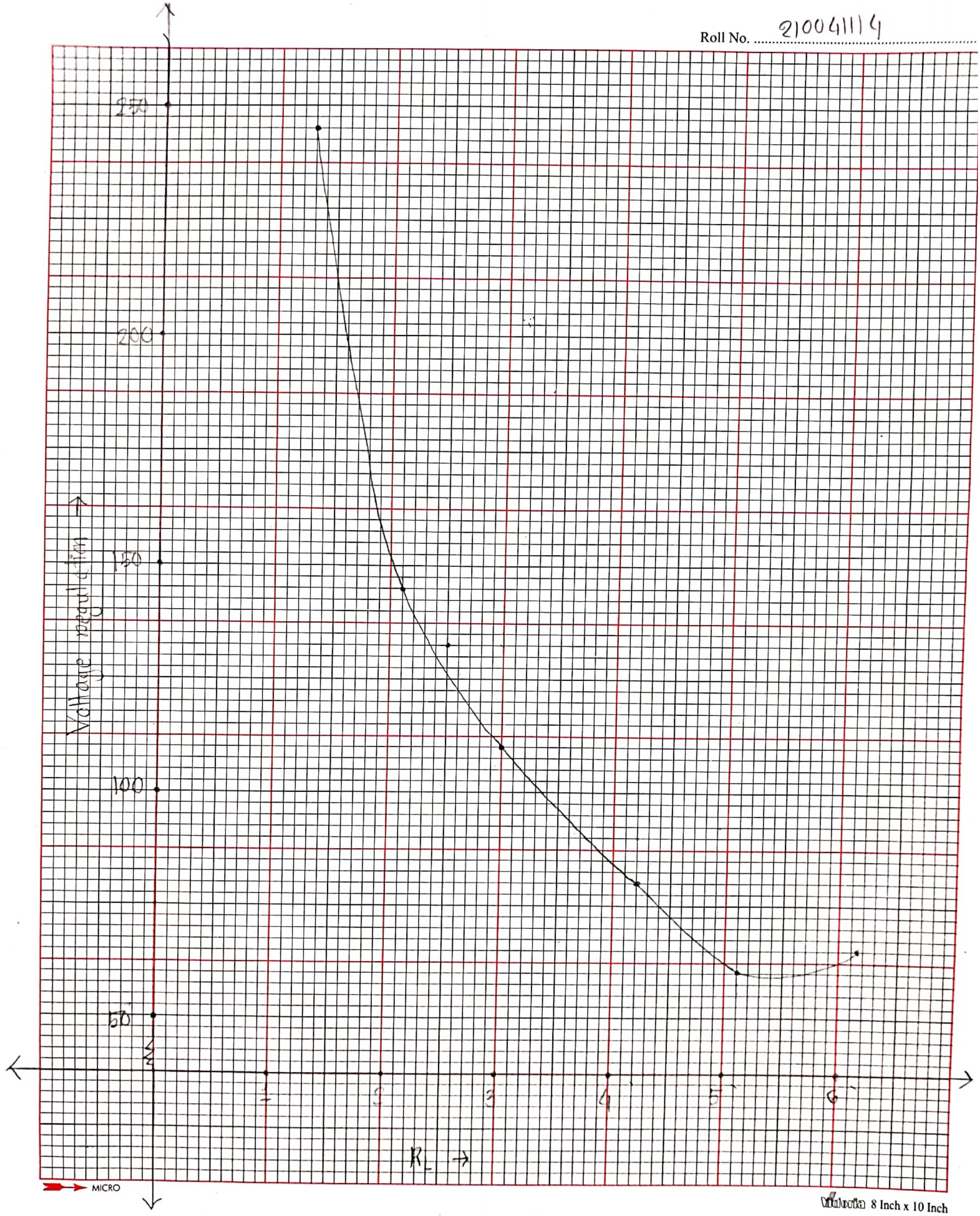

MICRO

Vidoria 8 Inch x 10 Inch

Result:

From the graphs we can conclude that maximum power transfer is used for observation 4.

Discussion and Q&A:

1) Each resistor was measured carefully before use.
2) The variable resistance was changed very slightly for every observation.
3) For max. power transfer $R_{TH} = R_L$ but in our collected data it isn't true. This might be due to some human error.

Maximum power transfer is used in circuits where optimum power transfer is required. It is also used in engines.

No I would not suggest using maximum power transfer in all cases. Because maximum power transfer does not mean maximum efficiency. So in circuits ~~we~~ where efficiency is important maximum power transfer should not be used.

<u>Efficiency of Power Transfer:</u> It is the percentage of Power we get from a system with regards to the supplied power. It is given by,

$$\eta = \frac{P_{OUT}}{P_{IN}} \times 100\%$$

<u>Efficiency of Voltage Regulation:</u> It is the percentage of voltage difference of V_{TH} and V_L with regards to V_L. It is given by,

$$\frac{V_{TH} - V_L}{V_L} \times 100\%$$